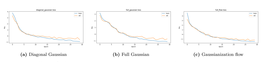
Exercise 3: Normalizing Flows for Stellar Labels and Uncertainties

This document shows the final plots for the three models I trained on the GALAH star spectra dataset. The models predict the stellar labels T_{eff}, $\log g$, and [Fe/H], and they also try to predict the uncertainty of those values using negative log-likelihood training with Jammy Flows.

I tested three versions of the model:

- **Diagonal Gaussian flow**: treats the uncertainty of each target separately.
- **Full Gaussian flow**: allows the three target uncertainties to be correlated.
- **Gaussianization flow**: a more flexible flow that can model non-Gaussian output distributions.

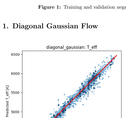
Training Curves



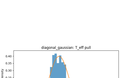
(a) Diagonal Gaussian (b) Full Gaussian (c) Gaussianization flow

Figure 1: Training and validation negative log-likelihood for the three flow models.

1. Diagonal Gaussian Flow

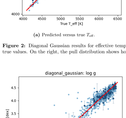


(a) Predicted versus true T_{eff}.



(b) Pull distribution for T_{eff}.

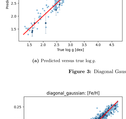
Figure 2: Diagonal Gaussian results for effective temperature. On the left, the predictions are compared with the true values. On the right, the pull distribution shows how reasonable the predicted uncertainties are.



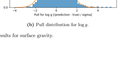
(a) Predicted versus true $\log g$.

(b) Pull distribution for $\log g$.

Figure 3: Diagonal Gaussian results for surface gravity.



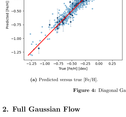
(a) Predicted versus true [Fe/H].



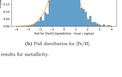
(b) Pull distribution for [Fe/H].

Figure 4: Diagonal Gaussian results for metallicity.

2. Full Gaussian Flow

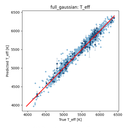


(a) Predicted versus true T_{eff}.

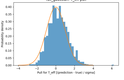


(b) Pull distribution for T_{eff}.

Figure 5: Full Gaussian results for effective temperature. This version can model correlations between the uncertainties of the three predicted labels.

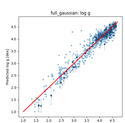


(a) Predicted versus true $\log g$.

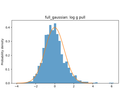


(b) Pull distribution for $\log g$.

Figure 6: Full Gaussian results for surface gravity.



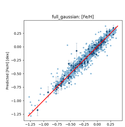
(a) Predicted versus true [Fe/H].



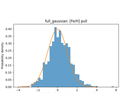
(b) Pull distribution for [Fe/H].

Figure 7: Full Gaussian results for metallicity.

3. Gaussianization Flow

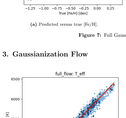


(a) Predicted versus true T_{eff}.



(b) Pull distribution for T_{eff}.

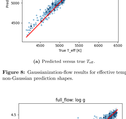
Figure 8: Gaussianization-flow results for effective temperature. This is the most flexible model, because it can learn non-Gaussian prediction shapes.



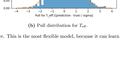
(a) Predicted versus true $\log g$.

(b) Pull distribution for $\log g$.

Figure 9: Gaussianization-flow results for surface gravity.



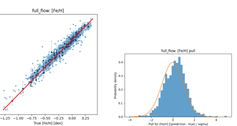
(a) Predicted versus true [Fe/H].



(b) Pull distribution for [Fe/H].

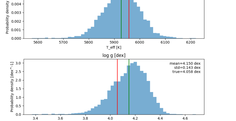
Figure 10: Gaussianization-flow results for metallicity.

4. Example Predicted PDFs from the Gaussianization Flow



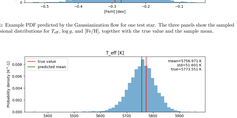
(a) Predicted T_{eff} for a star with $T_{\mathrm{eff}} = 5308\,\mathrm{K}$, $\log g = 4.2\,\mathrm{dex}$, and [Fe/H] = 0 dex.

(b) Predicted $\log g$ for a star with $T_{\mathrm{eff}} = 5308\,\mathrm{K}$, $\log g = 4.2\,\mathrm{dex}$, and [Fe/H] = 0 dex.

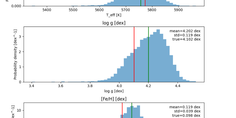


(c) Predicted [Fe/H] for a star with $T_{\mathrm{eff}} = 5308\,\mathrm{K}$, $\log g = 4.2\,\mathrm{dex}$, and [Fe/H] = 0 dex.

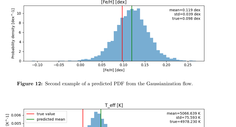
Figure 11: Example PDF predicted by the Gaussianization flow for one test star. The three panels show the sampled one-dimensional distributions for T_{eff}, $\log g$, and [Fe/H], together with the true value and the sample mean.



(a) Predicted T_{eff} for a star with $T_{\mathrm{eff}} = 5774\,\mathrm{K}$, $\log g = 4.4\,\mathrm{dex}$, and [Fe/H] = 0 dex.

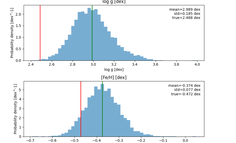


(b) Predicted $\log g$ for a star with $T_{\mathrm{eff}} = 5774\,\mathrm{K}$, $\log g = 4.4\,\mathrm{dex}$, and [Fe/H] = 0 dex.

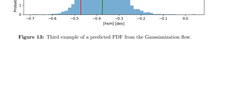


(c) Predicted [Fe/H] for a star with $T_{\mathrm{eff}} = 5774\,\mathrm{K}$, $\log g = 4.4\,\mathrm{dex}$, and [Fe/H] = 0 dex.

Figure 12: Second example of a predicted PDF from the Gaussianization flow.



(a) Predicted T_{eff} for a star with $T_{\mathrm{eff}} = 4978\,\mathrm{K}$, $\log g = 2.6\,\mathrm{dex}$, and [Fe/H] = −0.7 dex.



(b) Predicted $\log g$ for a star with $T_{\mathrm{eff}} = 4978\,\mathrm{K}$, $\log g = 2.6\,\mathrm{dex}$, and [Fe/H] = −0.7 dex.

(c) Predicted [Fe/H] for a star with $T_{\mathrm{eff}} = 4978\,\mathrm{K}$, $\log g = 2.6\,\mathrm{dex}$, and [Fe/H] = −0.7 dex.

Figure 13: Third example of a predicted PDF from the Gaussianization flow.